

- 1) A MOSFET is mounted on a FR4 PCB and has a total power loss of 5 W. The PCB is mounted on a heat sink which is surrounded by air. The ambient air temperature is 50°C. The thermal resistance $R_{th,jc}$ between the junction of the MOSFET (silicon die) and the top copper layer of the PCB is 1.5 °C/W. The thermal resistance $R_{th,ch}$ between the top copper layer of the PCB and the heat sink is 5 °C/W. The thermal resistance $R_{th,hs}$ of the heat sink is 2.5 °C/W. Please calculate the junction temperature of the MOSFET (temperature of MOSFET silicon). If the maximum junction temperature of the MOSFET is 150 °C, what is the maximum power dissipation that will not lead to any MOSFET damage with a margin of 10°C?

